

Network Fundamentals

Introduction

Theory + lab:

Lab:

Bachelor Electronics – ICT

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Learning objectives

1. The student understands how a computer network works and the individual components and protocols involved and can explain it.
2. The student is able to build a basic physical computer network
 - The student cables a physical network with the necessary network equipment and network clients
 - The student configures the clients and network equipment
 - The student tests the network
 - The student analyses the functioning of the network with various tools
3. The student is able to simulate a basic network
 - The student builds and configures a network in emulation software
 - The student analyses the functioning of the network and inspects the network packets
 - The student can explain the functioning of the protocols learned on the basis of the simulation
4. The student is able to demonstrate and explain the functioning of the computer network. The student has insight into the underlying functioning of the computer network and the protocols learned and can also answer technical questions about this
5. The student is able to clearly document their own knowledge acquisition and the functioning of the network

Course content

- Network applications
- Structure, components and types of networks
- Network stability
- Basic network protocols (ARP, DNS, DHCP, NAT)
- OSI (and TCP/IP) Model
 - Physical Layer (cabling/WiFi, modulation, encoding, bandwidth, etc.)
 - Data Link Layer (Ethernet)
 - Network Layer (IPv4, IPv6)
 - Transport Layer (TCP, UDP)
 - IP Addresses and subnetting
 - Application layer (FTP, TFTP, SMTP, etc.)
- Cisco IOS and client configuration

The lessons

Theory

- Theory lectures and demonstrations
- Take notes and ask questions

Lab

- Lab assignments building networks on physical devices
 - Executed during the lab sessions
- Exercises that give you a practical insight into the theory using virtual tools like Wireshark and packet tracer.
 - Executed during the lab sessions and at home

Evaluation

Theory

50%:

- Written exam, multiple choice + open questions
during exam period (June)

Lab

50%:

- **Permanent evaluation**
 - Lab assignments
 - Test subnetting
 - Theory exercise

The assessment activity for the individual course components (theory and laboratory) assesses individual learning objectives. Theory assesses learning objective 1, while the lab assesses the other learning objectives. If the student scores less than 6/20 for one of the teaching and learning activities (theory or lab), the total for the entire course can be a maximum of 9/20.

Retake

Theory 50%

- Written exam

Lab: 50%

- Practical exam

The assessment activity for the individual course components (theory and laboratory) assesses individual learning objectives. Theory assesses learning objective 1, while the lab assesses the other learning objectives. If the student scores less than 6/20 for one of the teaching and learning activities (theory or lab), the total for the entire course can be a maximum of 9/20.

Lab Assignments

- Groups of maximum 3 students
- During the lab, make your own notes, screenshots, photos to the lab assignment document → **logbook**
- When you finished the lab, **demonstrate** the lab to lector
- **Each student** should be able to explain/demonstrate each part of the lab during the demonstration
- When the lab is finished, each student **uploads** his/her own logbook on Canvas

IPv4 Subnetting test

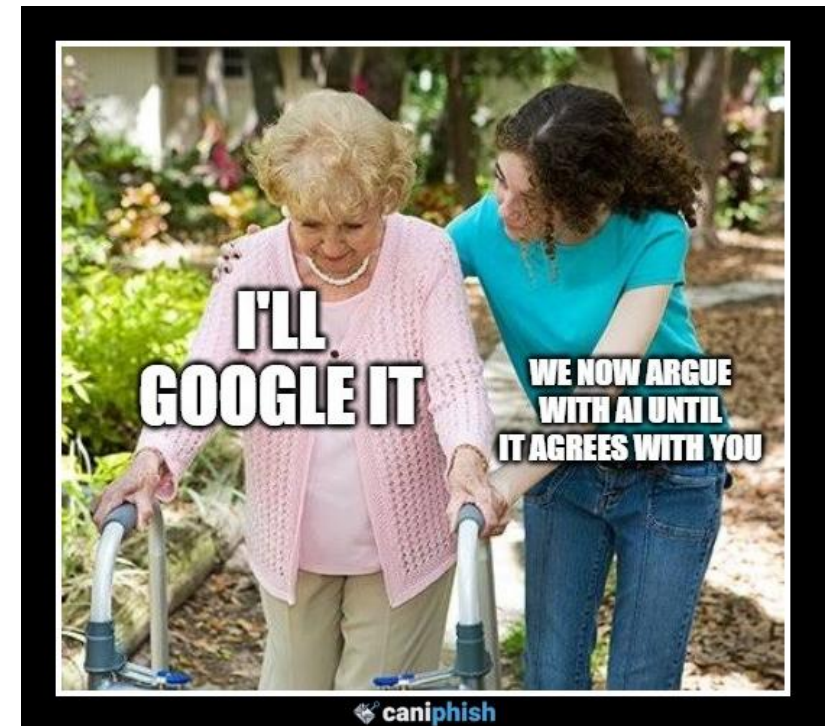
- Written test on paper
- During your normal lab session in the week of 4-8 May 2026
- More info during the subnetting theory lesson

Theory Exercise

- Test about one or more aspects of the exercises
- Wireshark/packet tracer
- During the test, you can use **your own notes, no internet**
- During your normal lab session in the week of 19-23 May 2025

The use of AI during this course

- You may use AI to support your assignments.
- Focus on achieving the objectives while demonstrating your ability to think critically.
- Document when and where you have used AI.
- Critically review and evaluate where AI has or has not contributed.
- Reflect critically on the AI output.



The use of AI during this course

- AI can be one of the tools to come to the course objectives, which is: Understanding and being able to implement Computer networks
- AI is not a tool that does your assignments and get your degree for you
- Don't become the future employee that can be replaced by AI (Copying and pasting the assignment into an AI tool can be done by anybody)
- Use AI as a tool to **elevate you**

